A Distribution Free Conditional Independence Test

with Application to Causal Discovery

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Framework

- Motivation and Introduction
- The Proposed Index
- Asymptotic Properties
- Simulation Results
- Application to Causal Discovery

Examples: Graphical Models

Let $X = (X_1, \ldots, X_p)^\top$ follows a multivariate Gaussian distribution. $\Sigma = \mathrm{Cov}(X)$. $\mathbf{Q} = \Sigma^{-1}$. Then

$$Q_{i,j} = 0 \Longleftrightarrow X_i \perp\!\!\!\perp X_j | X_{-ij}$$

Examples: Causal Inference

In observational studies, let $X = (X_1, \ldots, X_p)^\top$ be the covariates, T be the treatment variable, Y be the response. Then the Ignorability Condition says that:

$$Y \perp\!\!\!\perp T | X$$

Examples: Sufficient Dimension Reduction

Let $X = (X_1, \ldots, X_p)^\top$ be the covariates, Y be the response. Then the sufficient dimension reduction aims to find a $\beta \in \mathbb{R}^{p \times d}$, where $d << p$, such that:

$$Y \perp\!\!\!\perp X | \beta^\top X$$

Statistical Motivations

- Graphical Models
- Causal Inference
- Sufficient Dimension Reduction
- ....

Introduction: Conditional Independence

Let X, Y, Z be random variables. We are interested in whether X and Y are statistically independent when Z is observed. That is, we aim to test

$$H_0 : X \perp\!\!\!\perp Y \mid Z \quad \text{versus} \quad H_1 : \text{ Otherwise.}$$

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Fit the above two linear regressions, compute ϵ_1, ϵ_2. Then

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When (X, Y, Z) is jointly normal,

$$\mathrm{PCor}(X, Y) = 0 \Longleftrightarrow X \perp\!\!\!\perp Y \mid Z$$

Introduction: Linton et al. (1996)

Linton et al. (1996) proposes to use the test statistics based on

$$f(X, Y, Z)f(Z) - f(X, Z)f(Y, Z) = 0$$

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- Su and White (2008) measures the difference between the weighted Hellinger distance.

$$\int \left\{1 - \sqrt{\frac{f(x,y)f(y,z)}{f(x,y,z)f(z)}}\right\}^2 a(x,y,z)dF(x,y,z)$$

Introduction: Other Conditional Independence Tests

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- Our method?

Proposed Methodology

Assume X, Y and Z are univariate continuous random variables. Apply Rosenblatt transformation (Rosenblatt, 1952), let

$$U \stackrel{\text{def}}{=} F_{X|Z}(X \mid Z) \quad V \stackrel{\text{def}}{=} F_{Y|Z}(Y \mid Z) \quad W \stackrel{\text{def}}{=} F_Z(Z)$$

We observe that

$$X \perp\!\!\!\perp Y \mid Z \Longleftrightarrow U, V, W \text{ are mutually independent}$$

Mutual Independence

Proposition

Assume X, Y and Z are three univariate continuous random variables. Let $U \stackrel{\text{def}}{=} F_{X|Z}(X \mid Z)$, $V \stackrel{\text{def}}{=} F_{Y|Z}(Y \mid Z)$, $W \stackrel{\text{def}}{=} F_Z(Z)$, then $X \perp\!\!\!\perp Y \mid Z$ if and only if U, V and W are mutually independent.

Mutual Independence

Some remarks:

- We established an equivalence between the conditional independence and the mutual independence.
- For now, we consider the case where (X, Y, Z) are univariate and continuous.
- Thus any technique for testing mutual independence can be applied. For example, Chakraborty and Zhang (2019).
- We need to consider the intrinsic properties of (U, V, W).

Mutual Independence

Let $\omega(\cdot)$ be an arbitrary positive weight function and $\varphi_{U,V,W}(\cdot)$, $\varphi_U(\cdot)$, $\varphi_V(\cdot)$, and $\varphi_W(\cdot)$ be the characteristic functions of (U, V, W), U, V and W, respectively. Then

$$\begin{aligned} & U,\ V \text{ and } W \text{ are mutually independent} \\ \iff\ & \varphi_{U,V,W}(t,s,r) = \varphi_U(t)\varphi_V(s)\varphi_W(r) \text{ for all } t,s,r \in \mathbb{R} \\ \iff\ & \iiint \|\varphi_{U,V,W}(t,s,r) - \varphi_U(t)\varphi_V(s)\varphi_W(r)\|^2 \omega(t,s,r)dtdsdr = 0. \end{aligned}$$

Mutual Independence

By choosing $\omega(t, s, r)$ to be the joint pdf of three i.i.d. standard Cauchy random variables, the above integration has a closed form,

$$Ee^{-|U_1-U_2|-|V_1-V_2|-|W_1-W_2|} - 2Ee^{-|U_1-U_3|-|V_1-V_4|-|W_1-W_2|} + Ee^{-|U_1-U_2|}Ee^{-|V_1-V_2|}Ee^{-|W_1-W_2|}, \quad (1)$$

where (U_k, V_k, W_k), $k = 1, 2, 3, 4$, are four independent copies of (U, V, W).

Mutual Independence

Furthermore, with the fact that $U \perp\!\!\!\perp W$ and $V \perp\!\!\!\perp W$, we can further write it as

$$E\left\{S_U(U_1, U_2)S_V(V_1, V_2)e^{-|W_1-W_2|}\right\}, \tag{2}$$

where $S_U(U_1, U_2)$ and $S_V(V_1, V_2)$ are defined as

$$\begin{aligned} S_U &= E\left\{e^{-|U_1-U_2|} + e^{-|U_3-U_4|} - e^{-|U_1-U_3|} - e^{-|U_2-U_3|} \mid (U_1, U_2)\right\}, \\ S_V &= E\left\{e^{-|V_1-V_2|} + e^{-|V_3-V_4|} - e^{-|V_1-V_3|} - e^{-|V_2-V_3|} \mid (V_1, V_2)\right\}. \end{aligned}$$

Composite Conditional Index

Recall that U, V, and $W \sim U(0,1)$. With some further calculations, we now define an index $\rho(X, Y \mid Z)$ to measure the conditional dependence :

$$c_0 E\Big\{ \Big(e^{-|U_1-U_2|} + e^{-U_1} + e^{U_1-1} + e^{-U_2} + e^{U_2-1} + 2e^{-1} - 4\Big)$$
$$\Big(e^{-|V_1-V_2|} + e^{-V_1} + e^{V_1-1} + e^{-V_2} + e^{V_2-1} + 2e^{-1} - 4\Big) e^{-|W_1-W_2|}\Big\}, \quad (3)$$

where $c_0 = (13e^{-3} - 40e^{-2} + 13e^{-1})^{-1}$.

Composite Conditional Index

Theorem

Assume X, Y and Z are univariate continuous random variables. Let $U \stackrel{\text{def}}{=} F_{X|Z}(X \mid Z)$, $V \stackrel{\text{def}}{=} F_{Y|Z}(Y \mid Z)$, $W \stackrel{\text{def}}{=} F_Z(Z)$. The index $\rho(X, Y \mid Z)$ is defined in (5), then

(1) $\rho(X, Y \mid Z) \geq 0$, *where the equality holds if and only if* $X \perp\!\!\!\perp Y \mid Z$

(2) $\rho(X, Y \mid Z) \leq 1$, *where the equality holds if and only if* $U = V$ *or* $U + V = 1$.

(3) *The index* ρ *is conditionally symmetric, that is* $\rho(X, Y \mid Z) = \rho(Y, X \mid Z)$.

(4) *For any strictly monotone transformations* $m_1(\cdot)$, $m_2(\cdot)$ *and* $m_3(\cdot)$, $\rho(X, Y \mid Z) = \rho\{m_1(X), m_2(Y) \mid m_3(Z)\}$.

Asymptotic Properties

Sample version of $\widehat{\rho}(X, Y \mid Z)$:

$$c_0 n^{-2} \sum_{i,j} \left\{ \left(e^{-|\widehat{U}_i - \widehat{U}_j|} + e^{-\widehat{U}_i} + e^{\widehat{U}_i - 1} + e^{-\widehat{U}_j} + e^{\widehat{U}_j - 1} + 2e^{-1} - 4 \right) \right.$$
$$\left. \left(e^{-|\widehat{V}_i - \widehat{V}_j|} + e^{-\widehat{V}_i} + e^{\widehat{V}_i - 1} + e^{-\widehat{V}_j} + e^{\widehat{V}_j - 1} + 2e^{-1} - 4 \right) e^{-|\widehat{W}_i - \widehat{W}_j|} \right\}. \quad (4)$$

Advantages: It has the same asymptotic properties as if U, V, W are observed.
Computation complexity: $O(n^2)$.
In principal, when the conditional dependence of X and Y given Z is weak, the sample version of the index will also be closed to 0. Reversely, larger values of $\widehat{\rho}(X, Y \mid Z)$ will provide evidence against the conditional independence.

Null Hypothesis

We derive the limiting distribution of $\widehat{\rho}(X, Y \mid Z)$ under the null hypothesis in the following theorem.

Theorem

Assume the regularity conditions (C1)-(C4) hold. Under the null hypothesis,

$$n\widehat{\rho}(X, Y \mid Z) \xrightarrow{d} c_0 \sum_{j=1}^{\infty} \lambda_j \chi_j^2$$

where χ_j^2, $j = 1, 2, \ldots$ are independent chi-square random variables with one degree of freedom, and λ_js, $j = 1, 2, \ldots$ are positive numbers defined in the appendix, which does not depend on the joint distribution of (X, Y, Z).

Alternative Hypothesis

Next we study the power performance of our test under two kinds of alternative hypotheses. In general, when H_0 is false, the conditional independence violates. That is, under the global alternative, denoted by H_{1g}, we have

$$H_{1g}: \quad X \not\perp\!\!\!\perp Y \mid Z.$$

We also consider a sequence of local alternatives with the form

$$H_{1l}: \quad F_{X|Z}(x \mid Z = z) - F_{X|(Y,Z)}\{x \mid (Y = y, Z = z)\} = n^{-1/2}\ell(x, y, z).$$

In what follows, we discuss the asymptotic properties under the global alternative and local alternatives in the theorem below.

Global Alternative Hypothesis

Theorem

Assume the regularity conditions (C1)-(C4) hold. Under the global alternative hypothesis,

$$n^{1/2}\left\{\widehat{\rho}(X,Y\mid Z)-\rho(X,Y\mid Z)\right\}\xrightarrow{d}\mathcal{N}(0,\sigma_0^2),$$

where the variance $\sigma_0^2 \stackrel{\text{def}}{=} 4c_0^2\mathrm{var}(P_{1,i}+P_{2,i}+P_{3,i}+P_{4,i})$*, and* $P_{1,i}$*,* $P_{2,i}$*,* $P_{3,i}$ *and* $P_{4,i}$ *are defined in the appendix, respectively.*

Local Alternative Hypothesis

Theorem

Under the local alternative hypothesis,

$$n\widehat{\rho}(X, Y \mid Z) \xrightarrow{d} \iiint \|\zeta(t,s,r)\|^2 \, \omega(t,s,r) dtdsdr,$$

where $\zeta(t,s,r)$ *stands for a complex-valued Gaussian random process with mean function* $E\left[it\ell(X,Y,Z)e^{itU}\left\{e^{isV} - \varphi_V(s)\right\}e^{irW}\right]$ *and covariance function defined in the appendix, and* $\omega(t,s,r)$ *is the joint probability density function of three independent and identically distributed standard Cauchy random variables,*

Multivariate Case

Theorem

Assume $X_1, \ldots, X_p$, Y *and* Z *are univariate continuous random variables. Denote* $\widetilde{U}_1 = F_{X_1|Z}(X_1 \mid Z)$, $\widetilde{U}_k = F_{X_k|Z,X_1,\ldots,X_{k-1}}(X_k \mid Z, X_1, \ldots, X_{k-1})$ *for all* $k = 2, \ldots, p$, $V = F_{Y|Z}(Y \mid Z)$ *and* $W = F_Z(Z)$. *Then* $(X_1, \ldots, X_p) \perp\!\!\!\perp Y \mid Z$ *holds if and only if* $(\widetilde{U}_1, \ldots, \widetilde{U}_p, V, W)$ *are mutually independent.*

Multivariate Case

Now we are in the position to develop a new test for the mutual independence among $(\widetilde{U}_1, \ldots, \widetilde{U}_p, V, W)$, and equivalently, the test for the conditional independence $\mathbf{x} \perp\!\!\!\perp Y \mid Z$. Similar as the univariate case, with the fact that $(\widetilde{U}_1, \ldots, \widetilde{U}_p) \perp\!\!\!\perp W$, $V \perp\!\!\!\perp W$ and $\widetilde{U}_1, \ldots, \widetilde{U}_{p-1}$ and $\widetilde{U}_p$ are mutually independent, we define

$$T = E\left\{S_{\mathbf{u}}(\mathbf{u}_1, \mathbf{u}_2) S_V(V_1, V_2) e^{-|W_1 - W_2|}\right\},$$

where $\mathbf{u}_1, \mathbf{u}_2$ are two independent copies of $\mathbf{u} = (\widetilde{U}_1, \ldots, \widetilde{U}_p)^{\mathsf{T}}$, and $S_{\mathbf{u}}(\mathbf{u}_1, \mathbf{u}_2)$ is

$$E\left\{e^{-\|\mathbf{u}_1 - \mathbf{u}_2\|_1} + e^{-\|\mathbf{u}_3 - \mathbf{u}_4\|_1} - e^{-\|\mathbf{u}_1 - \mathbf{u}_3\|_1} - e^{-\|\mathbf{u}_2 - \mathbf{u}_3\|_1} \mid (\mathbf{u}_1, \mathbf{u}_2)\right\},$$

Multivariate Case

Recall that $\widetilde{U}_1, \ldots, \widetilde{U}_p$, V and W are all uniformly distributed, we can further simplify T as

$$E\Bigg[\Bigg\{e^{-\|\mathbf{u}_1-\mathbf{u}_2\|_1} - \sum_{i=1}^{2}\prod_{j=1}^{p}(2 - e^{-\widetilde{U}_{ij}} - e^{\widetilde{U}_{ij}-1}) + 2^p e^{-p}\Bigg\}$$
$$\left(e^{-|V_1-V_2|} + e^{-V_1} + e^{V_1-1} + e^{-V_2} + e^{V_2-1} + 2e^{-1} - 4\right) e^{-|W_1-W_2|}\Bigg],$$

where $\widetilde{U}_{ij}$ is the jth element of $\mathbf{u}_i$, $i = 1, 2$, $j = 1, \ldots, p$. Then T is nonnegative and equals zero if and only if $\mathbf{x} \perp\!\!\!\perp Y \mid Z$ and it can be used to measure the conditional dependence between $\mathbf{x}$ and Y given Z.

Discrete Case

Apply transformations in Brockwell (2007) to obtain U and V. Specifically, define $F_{X|Z}(x \mid z) = P(X \leq x \mid Z = z)$, $F_{X|Z}(x- \mid z) = P(X < x \mid Z = z)$, $F_{Y|Z}(y \mid z) = P(Y \leq y \mid Z = z)$, and $F_{Y|Z}(y- \mid z) = P(Y < y \mid Z = z)$, let U_X and U_Y be two independent and identically distributed $U(0,1)$ random variables, and do the transformations as follows,

$$\begin{aligned} U &= (1 - U_X)F_{X|Z}(X- \mid Z) + U_X F_{X|Z}(X \mid Z), \\ V &= (1 - U_Y)F_{Y|Z}(Y- \mid Z) + U_Y F_{Y|Z}(Y \mid Z). \end{aligned}$$

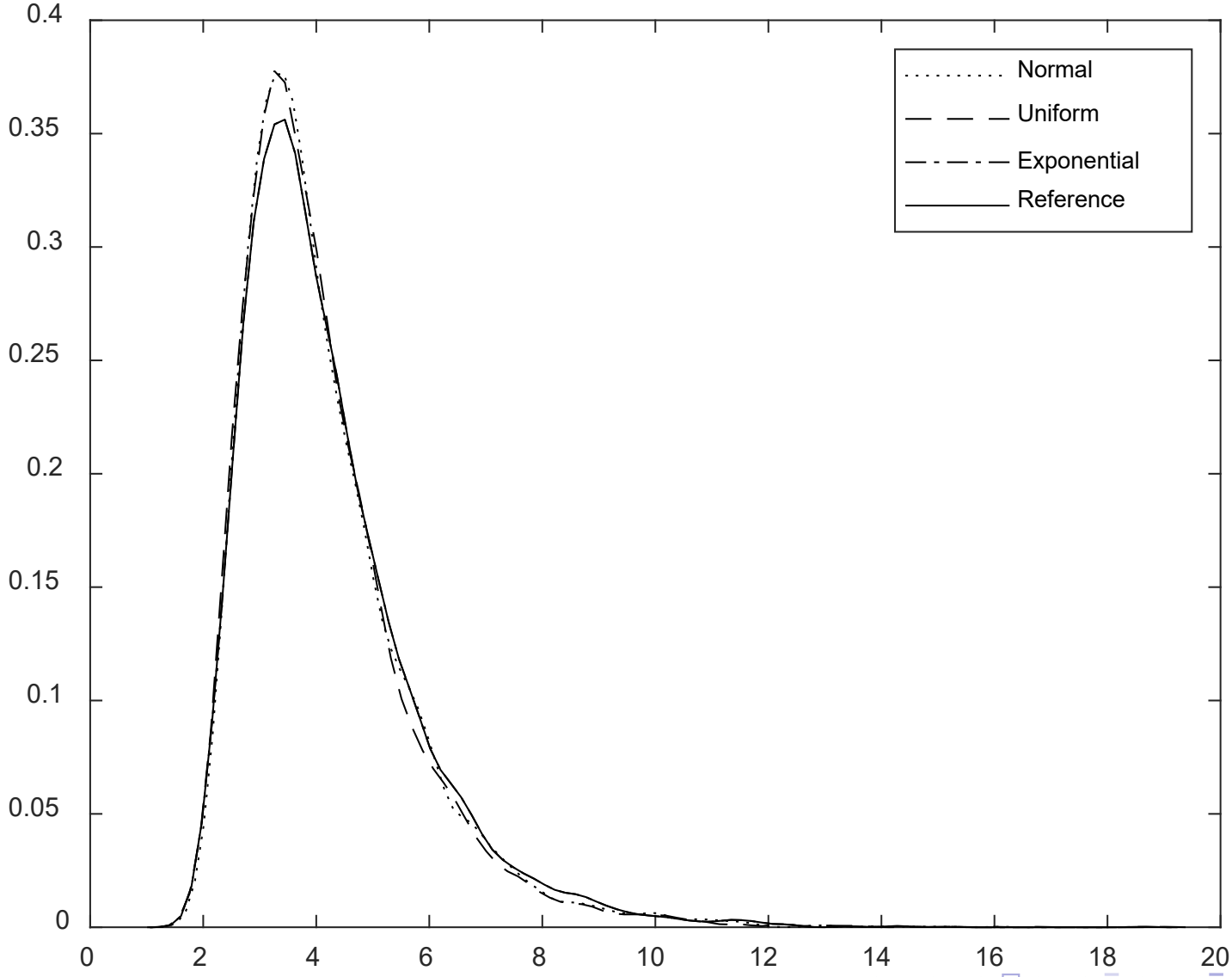
According to Brockwell (2007), both U and V are uniformly distributed on $(0,1)$. In addition, $U \perp\!\!\!\perp Z$ and $V \perp\!\!\!\perp Z$.

Discrete Case

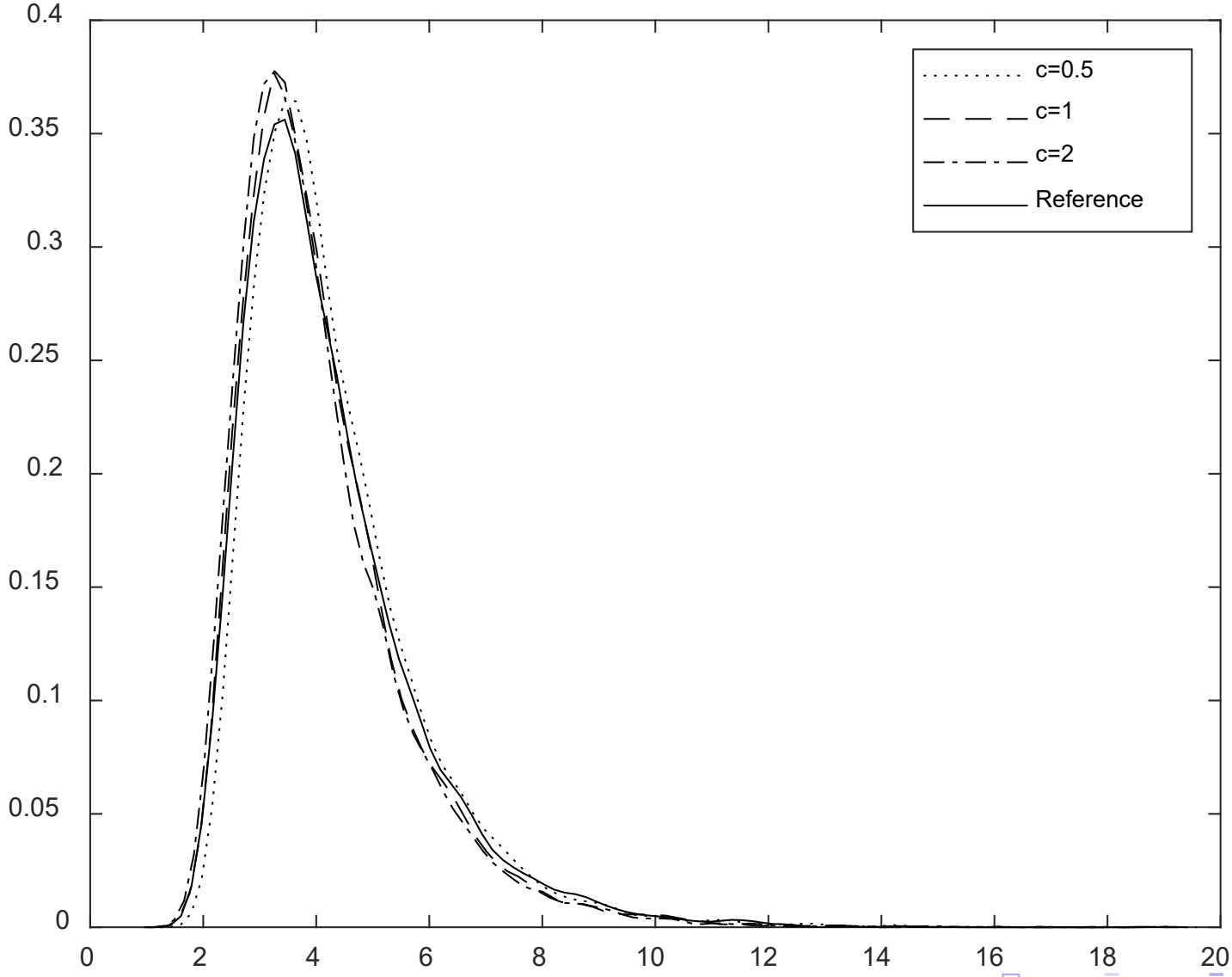
Proposition

Assume X, Y and Z are three univariate discrete random variables. Let $U \stackrel{\text{def}}{=} (1 - U_X)F_{X|Z}(X- \mid Z) + U_X F_{X|Z}(X \mid Z)$, $V \stackrel{\text{def}}{=} (1 - U_Y)F_{Y|Z}(Y- \mid Z) + U_Y F_{Y|Z}(Y \mid Z)$, *where U_X and U_Y are two independent standard uniformly distributed random variables. Then $X \perp\!\!\!\perp Y \mid Z$ if and only if U, V and Z are mutually independent.*

Distribution Free: Different Distribution



Distribution Free: Different Bandwidth



Simulations

1. $X_1 \sim N(0,1)$, $X_2 \sim N(0,1)$, $Z \sim N(0,1)$. $X = X_1 + Z$, $Y = X_2 + Z$. Thus X and Y are conditionally independent given Z.
2. $X_1 \sim N(0,1)$, $Z \sim N(0,1)$, $X = X_1 + Z$, $Y = X_1^2 + Z$.
3. $X_1 \sim N(0,1)$, $Z \sim N(0,1)$, $X = X_1 + Z$, $Y = X_1^4 + Z$.
4. $X_1 \sim N(0,1)$, $X_2 \sim N(0,1)$, $Z \sim N(0,1)$, $X = X_1 + Z$, $Y = X_1^2 + X_2 + Z$.
5. $X_1 \sim N(0,1)$, $X_2 \sim N(0,1)$, $Z \sim N(0,1)$, $X = X_1 + Z$, $Y = X_1^4 + X_2 + Z$.
6. $X_1 \sim N(0,1)$, $Z \sim N(0,1)$, $X = X_1 + Z$, $Y = 0.5 * \sin(\pi X_1) + Z$.

Simulations

Samples	Signi.	Test	M1	M2	M3	M4	M5	M6
50	0.05	CCI	0.056	1.000	1.000	0.676	0.866	0.572
		CDC	0.050	0.886	0.934	0.536	0.868	0.338
		CMI	0.048	0.380	0.526	0.316	0.518	0.070
		KCI	0.038	0.884	0.832	0.330	0.292	0.250
	0.1	CCI	0.098	1.000	1.000	0.798	0.934	0.712
		CDC	0.134	0.970	0.982	0.728	0.954	0.562
		CMI	0.088	0.484	0.598	0.420	0.582	0.132
		KCI	0.088	0.968	0.938	0.492	0.506	0.344
100	0.05	CCI	0.048	1.000	1.000	0.990	1.000	0.978
		CDC	0.066	0.998	0.996	0.880	0.998	0.694
		CMI	0.054	0.402	0.542	0.350	0.544	0.070
		KCI	0.040	1.000	0.998	0.790	0.842	0.444
	0.1	CCI	0.112	1.000	1.000	0.998	1.000	0.998
		CDC	0.158	1.000	1.000	0.980	1.000	0.834
		CMI	0.098	0.496	0.610	0.422	0.604	0.124
		KCI	0.088	1.000	1.000	0.878	0.926	0.598

Simulations

7. $X_1,\ X_2,\ Z_1,\ Z_2 \sim N(0,1)$, $X = X_1 + Z_1 + Z_2$, $Y = X_2 + Z_1 + Z_2$, $Z = (Z_1, Z_2)$. Thus X and Y are conditionally independent given Z.
8. $X_1,\ X_2,\ Z_1,\ Z_2 \sim N(0,1)$, $X = X_1^2 + Z_1 + Z_2$, $Y = \log(X_1 + 10) + Z_1 + Z_2$, $Z = (Z_1, Z_2)$.
9. $X_1,\ X_2,\ Z_1,\ Z_2 \sim N(0,1)$, $X = \tanh(X_1) + Z_1 + Z_2$, $Y = \log(X_1^2 + 10) + Z_1 + Z_2$, $Z = (Z_1, Z_2)$.
10. $X_1,\ X_2,\ Z_1,\ Z_2 \sim N(0,1)$, $X = X_1^2 + Z_1 + Z_2$, $Y = \log(X_1 Z_1 + 10) + Z_1 + Z_2$, $Z = (Z_1, Z_2)$.
11. $X_1,\ X_2,\ Z_1,\ Z_2 \sim N(0,1)$, $X = X_1 + Z_1 + Z_2$, $Y = \sin(X_1 Z_1) + Z_1 + Z_2$, $Z = (Z_1, Z_2)$.
12. $X_1,\ X_2,\ Z_1,\ Z_2 \sim N(0,1)$, $X = \log(X_1 Z_1 + 10) + Z_1 + Z_2$, $Y = \exp(X_1 Z_2) + Z_1 + Z_2$, $Z = (Z_1, Z_2)$.

Simulations

Samples	Signi.	Test	M7	M8	M9	M10	M11	M12
50	0.05	CCI	0.046	0.672	0.906	0.686	0.440	0.788
		CDC	0.052	0.408	0.850	0.054	0.128	0.134
		CMI	0.072	0.426	0.192	0.392	0.118	0.300
		KCI	0.046	0.030	0.088	0.026	0.662	0.248
	0.1	CCI	0.092	0.792	0.948	0.798	0.582	0.874
		CDC	0.126	0.670	0.976	0.194	0.298	0.264
		CMI	0.120	0.506	0.292	0.500	0.210	0.386
		KCI	0.098	0.092	0.190	0.074	0.820	0.492
100	0.05	CCI	0.048	0.936	0.998	0.936	0.664	0.988
		CDC	0.084	0.890	1.000	0.920	0.972	0.306
		CMI	0.044	0.412	0.164	0.392	0.126	0.300
		KCI	0.044	0.038	0.172	0.028	0.990	0.358
	0.1	CCI	0.104	0.958	1.000	0.966	0.766	0.996
		CDC	0.168	0.978	1.000	0.986	0.992	0.456
		CMI	0.128	0.486	0.240	0.460	0.194	0.390
		KCI	0.090	0.092	0.358	0.108	0.998	0.608

Applications to Causal Discovery

One of the most important application for conditional independence is causal discovery in directed acyclic graphs $G = (V, E)$, which consists of a set of nodes $V = \{1, 2, \ldots, p\}$ corresponding to a random vector $X = (X_1, \ldots, X_p) \in \mathbb{R}^p$, and a set of edges $E \subset V \times V$ such that there are no directed cycles. Two vertices X_1 and X_2 are *d-separated* by a subset of vertices S if every path between them is blocked by S. Denote the joint distribution of X by $P(X)$. The joint distribution is said to be *faithful* with respect to a graph G if and only if for any $i, j \in V$, and any subset $S \subset V$,

$X_i \perp\!\!\!\perp X_j | \{X_r : r \in S\} \Leftrightarrow$ node i and node j are d-separated by the set S

Causal Discovery

One of the most famous algorithm for recovering the graphs satisfying the *faithfulness* assumption is the PC algorithm (Spirtes et al., 2000; Kalisch and Bühlmann, 2007). The algorithm could cover the graph up to its *Markov equivalence class*, which are sets of graphs that entail the same set of (conditional) independences.

Causal Discovery

Following Kalisch and Bühlmann (2007), we consider the directed acyclic graph with 5 nodes, i.e., $X = (X_1, \ldots, X_5)$, and only allow directed edge from X_i and X_j for $i < j$. Denote the adjacency matrix A. The existence of the edge follows a Bernoulli distribution, and we set $P(A_{i,j} = 1) = 0.4$, for $i < j$. When $A_{i,j} = 1$, we replace $A_{i,j}$ with independent realizations of a uniform $U(0.1, 1)$ random variable. The value of the first random variable X_1 is randomly sampled from some distribution $\widetilde{P}$. Specifically,

$$\epsilon_1 \sim \widetilde{P}, \text{ and } X_1 = \epsilon_1$$

The value of the next nodes are given by

$$\epsilon_j \sim \widetilde{P}, \text{ and } X_j = \sum_{k=1}^{j-1} A_{j,k} X_k + \epsilon_j$$

All random errors $\epsilon_1, \ldots, \epsilon_p$ are independently sampled from the distribution $\widetilde{P}$. We consider two scenarios where $\widetilde{P}$ is either uniform $U(0, 1)$ or normal $N(0, 1)$ distribution.

Causal Discovery

Table: The true positive rate and the false positive rate for the causal discovery of the directed acyclic graph with different test method. CCI is the proposed test, when $\widetilde{P} \sim N(0,1)$. PCOR stands for the partial correlation (Kalisch and Bühlmann, 2007). CMI is the conditional mutual information (Scutari, 2010) while KCI is the kernel based conditional independence test (Zhang et al., 2011).

Test	Criterion	50	100	200	300
CCI	TPR	0.555	0.658	0.734	0.789
	FPR	0.117	0.112	0.107	0.103
PCOR	TPR	0.479	0.489	0.546	0.589
	FPR	0.101	0.110	0.130	0.135
CMI	TPR	0.472	0.530	0.604	0.590
	FPR	0.097	0.127	0.143	0.150
KCI	TPR	0.360	0.516	0.592	0.634
	FPR	0.072	0.135	0.144	0.168

Causal Discovery

Table: The true positive rate and the false positive rate for the causal discovery of the directed acyclic graph with different test method. CCI is the proposed test, when $\widetilde{P} \sim U(0,1)$. PCOR stands for the partial correlation (Kalisch and Bühlmann, 2007). CMI is the conditional mutual information (Scutari, 2010) while KCI is the kernel based conditional independence test (Zhang et al., 2011).

Test	Criterion	50	100	200	300
CCI	TPR	0.468	0.587	0.734	0.736
	FPR	0.070	0.099	0.095	0.113
PCOR	TPR	0.469	0.526	0.588	0.545
	FPR	0.111	0.129	0.140	0.140
CMI	TPR	0.497	0.568	0.566	0.633
	FPR	0.103	0.127	0.138	0.149
KCI	TPR	0.386	0.458	0.523	0.564
	FPR	0.082	0.099	0.123	0.122

Real Data Analysis

We analyse a real data set originally from the National Institute of Diabetes and Digestive and Kidney Diseases (Smith et al., 1988). We are interested in the causal structural of five variables: Age, Body Mass Index (BMI), 2-Hour Serum Insulin (SI), Plasma Glucose Concentration (GLU) and Diastolic Blood Pressure (DBP). After removing the missing data, we obtain $n = 392$ samples, which is the same with Chakraborty and Zhang (2019).

Real Data Analysis

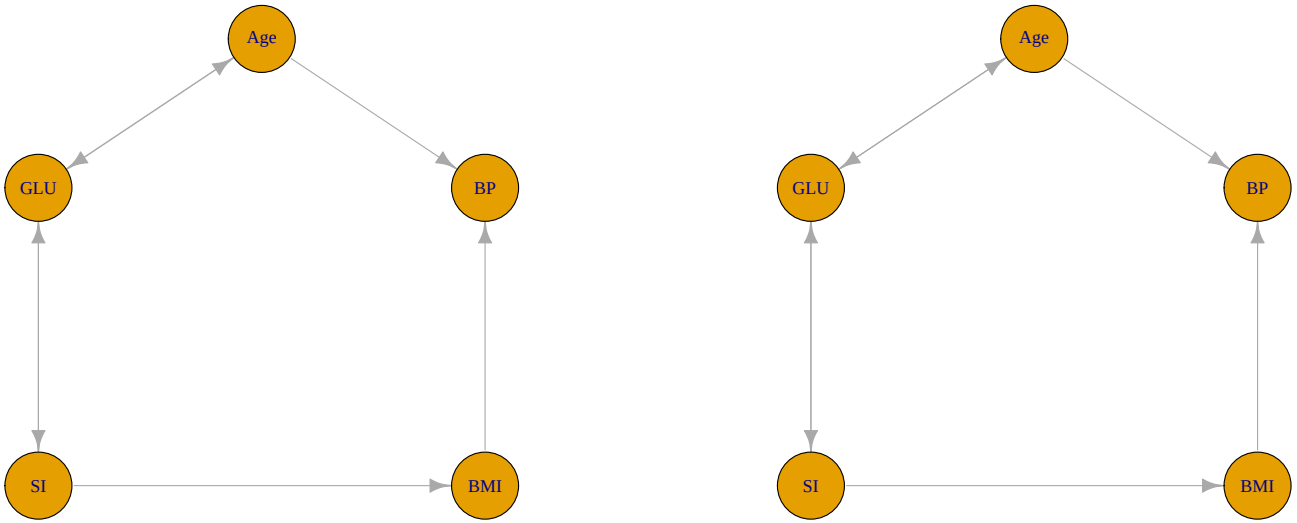


Figure: The estimated causal structure of the five variables by using proposed composite conditional index (left), partial correlation (right).

Real Data Analysis

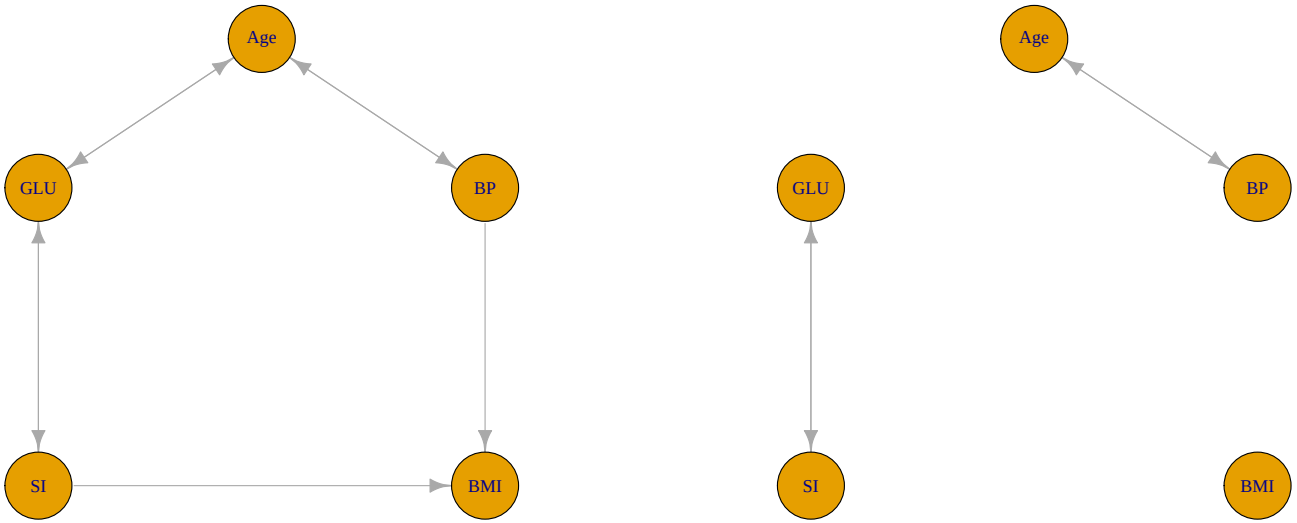


Figure: The estimated causal structure of the five variables by using conditional mutual information (left) and kernel based conditional independence test (right).

Conclusion

In this paper we developed a new conditional independence measure. It could identify the conditional dependence consistently and is computational fast. Numerical simulations indicates that the test is more powerful compared to other existing methods. We also applied the proposed test to a causal discovery in directed acyclic graphs.

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